

Solutions to MP204 Exam, Spring 2015

(1)

P.1

This problem tests the knowledge of how Maxwell's eqs arise from their original integral forms to the more familiar differential forms.

(a) Maxwell's First Eq'n (AKA Gauss' Law) says

$$\oint_S \vec{E} \cdot d\vec{S} = Q_S / \epsilon_0$$

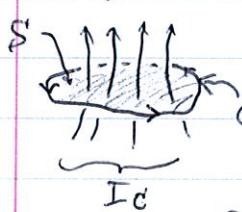
S here is a closed surface, and Q_S denotes the total electric charge [5 pts]

inside S . By def'n, the charge density ρ is that fraction which says how much charge dQ is contained within an infinitesimal volume dV , via $dQ = \rho dV$ if we integrate this over the region V enclosed within S , that gives Q_S , so $Q_S = \int_V \rho dV$. [5 pts]

In Maxwell's Fourth Eq'n (AKA the extended Ampère law), we have

$$\oint_C \vec{B} \cdot d\vec{r} = \mu_0 I_C + \mu_0 \epsilon_0 \int_S \frac{\partial \vec{E}}{\partial t} \cdot d\vec{S}, \text{ where } C \text{ is a closed curve}$$

and S the surface bounded by it, I_C is the total current which



C encloses, i.e. the total current flowing through S . [5 pts]

Def'n of the current density $\vec{J}$ tells us that the current dI flowing through a small area element $d\vec{S}$ is $dI = \vec{J} \cdot d\vec{S}$,

so since I_C is the total current flowing through S , it's obtained

by integrating dI over S , i.e. $I_C = \int_S \vec{J} \cdot d\vec{S}$ [5 pts]

(b) Now, to use (a) to rewrite the four eq's:

$$\oint_S \vec{E} \cdot d\vec{S} - \int_V \frac{\rho}{\epsilon_0} dV = 0, \quad \oint_S \vec{B} \cdot d\vec{S} = 0$$

$$\oint_C \vec{E} \cdot d\vec{r} + \int_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S} = 0, \quad \oint_C \vec{B} \cdot d\vec{r} - \int_S (\mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}) \cdot d\vec{S} = 0$$

Now we can use the two big theorems of vector calculus, Gauss' Theorem

Reynolds and Stokes' Theorem. The first says that for any vector field $\vec{A}$,

$$\oint_S \vec{A} \cdot d\vec{S} = \int_V (\vec{\nabla} \cdot \vec{A}) dV, \text{ where } S \text{ is a closed surface and } V \text{ the volume [5 pts]}$$

within. Using this, we get

$$\oint_S \vec{E} \cdot d\vec{S} - \int_V \frac{\rho}{\epsilon_0} dV = \int_V (\vec{\nabla} \cdot \vec{E} - \frac{\rho}{\epsilon_0}) dV = 0$$

$$\oint_S \vec{B} \cdot d\vec{S} = \int_V (\vec{\nabla} \cdot \vec{B}) dV = 0$$

Both of these must hold for any volume V whatsoever, which can only happen if their integrands vanish identically. Thus, we find

$$\vec{\nabla} \cdot \vec{E} = \rho / \epsilon_0 \quad \text{and} \quad \vec{\nabla} \cdot \vec{B} = 0$$

[5 pts each]

(2)

Stokes' Theorem says that if C is a closed curve enclosing a surface S , then $\oint_C \vec{A} \cdot d\vec{r} = \int_S (\vec{\nabla} \times \vec{A}) \cdot d\vec{S}$ for any vector field $\vec{A}$. This allows us to write

$$\oint_C \vec{E} \cdot d\vec{r} + \int_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S} = \int_S (\vec{\nabla} \times \vec{E} + \frac{\partial \vec{B}}{\partial t}) \cdot d\vec{S} = 0$$

$$\oint_C \vec{B} \cdot d\vec{r} - \int_S (\mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}) \cdot d\vec{S} = \int_S (\vec{\nabla} \times \vec{B} - \mu_0 \vec{J} - \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}) \cdot d\vec{S} = 0.$$

Again, this holds for all S , and so the integrands must vanish, giving

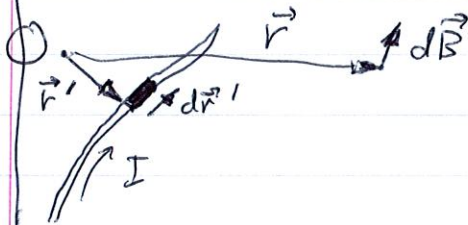
$$\boxed{\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad \text{and} \quad \vec{\nabla} \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}}$$

[5pts each]

P.2

The explanation in (b) was done in a tutorial, so the students have seen it before. It shouldn't prove too hard. But first, they have to understand the B-S Law.

(a) The B-S Law gives us the contribution to the total magnetic field from each little bit of wire in a loop. More specifically, a small bit of wire at position $\vec{r}'$, of length $|d\vec{r}'|$ has a current I flowing through it. We take the direction of $d\vec{r}'$ to be that of the current flow. Now we look



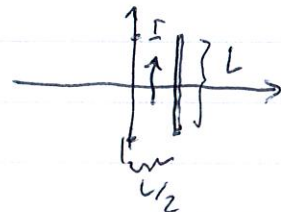
to see what the magnetic field due to this little bit of current is at the point $\vec{r}$. The B-S Law tells us that

$$d\vec{B} = \frac{\mu_0 I}{4\pi} \frac{d\vec{r}' \times (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3}$$

is that contribution.

[20pts]

(b) This problem can be made much easier by using symmetry, namely, each little segment of wire along one side contributes equally to the magnetic field at the center. Thus, if we just compute the field at the origin for the wire shown to the right, and multiply by 4, we get the answer. So let's do that.



$\vec{r} = \vec{0}$ is the point we're looking at, and we see that all parts of the wire are described by $\vec{r}' = \frac{L}{2} \hat{i} + y \hat{j}$, where $-\frac{L}{2} \leq y \leq \frac{L}{2}$. Thus, $d\vec{r}' = dy \hat{j}$, and so

$$d\vec{r}' \times (\vec{r} - \vec{r}') = dy \hat{j} \times (-\frac{L}{2} \hat{i} - y \hat{j}) = \frac{L}{2} dy \hat{k}.$$

(3)

Therefore,

$$d\vec{B}_{\text{center}} = \mu_0 I \times \frac{1}{4\pi} \cdot \frac{L/2 dy}{|-\frac{L}{2} + y|^3} = \frac{\mu_0 I L dy}{2\pi (y^2 + L^2/4)^{3/2}}$$

Therefore,

$$\vec{B}_{\text{center}} = \frac{\mu_0 I L}{2\pi} \int_{-L/2}^{L/2} \frac{dy}{(y^2 + L^2/4)^{3/2}} = \frac{\mu_0 I L}{\pi} \int_0^{L/2} \frac{dy}{(y^2 + L^2/4)^{3/2}}$$

because the integrand is an even function of y . Making substitution $y = \frac{L}{2} \tan \theta$,

we get

$$\vec{B}_{\text{center}} = \frac{\mu_0 I L}{\pi} \int_0^{\pi/4} \frac{\frac{L}{2} \sec^2 \theta}{(\frac{L^2}{4} \sec^2 \theta)^{3/2}} d\theta = \frac{4\mu_0 I}{\pi L} \int_0^{\pi/4} \cos \theta d\theta$$

$$= \frac{4\mu_0 I}{\pi L} \sin \theta \Big|_0^{\pi/4} = \frac{4\mu_0 I}{\pi L} \frac{1}{\sqrt{2}} = \boxed{\frac{2\sqrt{2}\mu_0 I L}{\pi L}} \quad [30 \text{ pts}]$$

P.3

If they've done P.1, then (a) should be immediate. If they haven't, then at least P.1 will give them a hint as to what Maxwell's Eqs are if they're forgotten.

~~(a) Remember "vacuum" means $\rho = 0, \vec{J} = 0$, so we have $\vec{E} = 0, \vec{B} = 0, \vec{\nabla} \cdot \vec{E} = 0, \vec{\nabla} \cdot \vec{B} = 0, \vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}, \vec{\nabla} \times \vec{B} = \mu_0 \frac{\partial \vec{E}}{\partial t}$.~~

(a) The way we calculate $\vec{E}$ & $\vec{B}$ from the potentials Φ and $\vec{A}$ is just

$$\boxed{\vec{E} = -\vec{\nabla} \Phi - \frac{\partial \vec{A}}{\partial t}, \quad \vec{B} = \vec{\nabla} \times \vec{A}}$$

[5 pts each]

$\vec{\nabla} \cdot \vec{B} = 0$ is immediate, because $\vec{\nabla} \cdot \vec{B} = \vec{\nabla} \cdot (\vec{\nabla} \times \vec{A})$, and the divergence of a curl always vanishes. Thus $\vec{\nabla} \cdot \vec{B} = 0$ is always satisfied.

[5 pts]

$\vec{\nabla} \times \vec{E} = \vec{\nabla} \times (-\vec{\nabla} \Phi - \frac{\partial \vec{A}}{\partial t})$, and since the curl of a gradient always vanishes, $\vec{\nabla} \times \vec{E} = -\vec{\nabla} \times \frac{\partial \vec{A}}{\partial t}$. We can always compute derivatives in any order, so we can compute the curl before the time derivative, i.e. $\vec{\nabla} \times \vec{E} = -\frac{\partial}{\partial t} (\vec{\nabla} \times \vec{A})$. But $\vec{\nabla} \times \vec{A} = \vec{B}$, so $\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$, and so Maxwell's third Eq is satisfied.

[5 pts]

(b) "Vacuum" = no charge or current density, i.e. $\boxed{\rho = 0, \vec{J} = 0}$

[5 pts each]

(c) If we put $\vec{E} = -\vec{\nabla} \Phi - \frac{\partial \vec{A}}{\partial t}$ into Maxwell's first Eq, we get

$$\vec{\nabla} \cdot \vec{E} = \vec{\nabla} \cdot (-\vec{\nabla} \Phi - \frac{\partial \vec{A}}{\partial t}) = -\nabla^2 \Phi - \vec{\nabla} \cdot \frac{\partial \vec{A}}{\partial t} = 0.$$

We can switch the order of the derivatives in the second term to get $\frac{\partial}{\partial t} (\vec{\nabla} \cdot \vec{A})$; however,

4)

The Lorentz condition says that $\vec{\nabla} \cdot \vec{A} = -\frac{1}{c^2} \frac{\partial \Phi}{\partial t}$, so $\frac{\partial}{\partial t} (\vec{\nabla} \cdot \vec{A}) = -\frac{1}{c^2} \frac{\partial^2 \Phi}{\partial t^2}$.

Plugging this back into Maxwell's First eqn ~~gives~~ ~~gives~~ gives

$$-\nabla^2 \Phi + \frac{1}{c^2} \frac{\partial^2 \Phi}{\partial t^2} = 0 \Rightarrow \boxed{\nabla^2 \Phi = \frac{1}{c^2} \frac{\partial^2 \Phi}{\partial t^2}} \quad [10 \text{ pts}]$$

This is a wave eq'n with speed c , so Φ is indeed described by such a wave.

For the vector potential, we take the Fourth Maxwell eqn:

$$\vec{\nabla} \times \vec{B} = \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \Rightarrow \vec{\nabla} \times (\vec{\nabla} \times \vec{A}) = \mu_0 \epsilon_0 \frac{\partial}{\partial t} (-\vec{\nabla} \Phi - \frac{\partial \vec{A}}{\partial t})$$

$$\Rightarrow \vec{\nabla} (\vec{\nabla} \cdot \vec{A}) - \nabla^2 \vec{A} = -\mu_0 \epsilon_0 \frac{\partial}{\partial t} \vec{\nabla} \Phi - \mu_0 \epsilon_0 \frac{\partial^2 \vec{A}}{\partial t^2}$$

$$\Rightarrow \nabla^2 \vec{A} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{A}}{\partial t^2} + \vec{\nabla} (\vec{\nabla} \cdot \vec{A} + \mu_0 \epsilon_0 \frac{\partial \Phi}{\partial t})$$

Here we have, once again, used the fact that partial derivatives can be done in any order. Now, $\mu_0 \epsilon_0 = 1/c^2$, so the term in the brackets vanishes, since it just becomes the Lorentz condition. Thus, we find

$$\boxed{\nabla^2 \vec{A} = \frac{1}{c^2} \frac{\partial^2 \vec{A}}{\partial t^2}} \quad [10 \text{ pts}]$$

and so the vector potential also satisfies a wave eq'n.